JIRA SPRINT & USER STORIES DOCUMENT

Healthcare Insurance Data Pipeline Project

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1. Project Overview

This project aims to build a data pipeline for a healthcare insurance company using Big Data technologies. The pipeline will ingest, clean, process, and analyze data to generate insights that help improve revenue, understand customer behavior, and optimize insurance offerings.

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2. Sprint Planning

Sprint Duration: 2 Weeks

Sprint 1: Documentation & Design (Week 1)

Goal: Define project structure and architecture

Activities:

- Analyze project requirements
- Create Requirements Specification Document (SRS)
- Create Solution Design Document
- Design database schema (Redshift tables)
- Identify business use cases
- Create Jira user stories

Sprint 2: Implementation & Testing (Week 2)

Goal: Build and validate the data pipeline

Activities:

- Upload datasets to AWS S3
- Perform data cleaning using PySpark
- Load cleaned data into Redshift
- Create output tables for use cases
- Write SQL queries for analytics
- Perform testing and validation
- Capture screenshots for submission
- Push code and documents to GitHub

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3. Jira Epics and User Stories

🎯 EPIC 1: Data Ingestion & Cleaning

User Story 1

As a Data Engineer, I want to upload raw CSV data to AWS S3 so that it can be processed for analysis.

Tasks:

- Create S3 bucket
- Upload input datasets
- Verify data upload

User Story 2

As a Data Engineer, I want to clean data using PySpark so that the dataset is accurate and reliable.

Tasks:

- Identify null values

- Count null values per column
- Replace null values with "NA"
- Remove duplicate records

🎯 EPIC 2: Data Processing & Storage

User Story 3

As a Data Engineer, I want to transform raw datasets so that they are structured for analysis.

Tasks:

- Apply schema to datasets
- Perform data type conversions
- Validate cleaned data

User Story 4

As a Data Engineer, I want to load cleaned data into Redshift so that it can be used for querying.

Tasks:

- Create Redshift schema
- Create tables with PK/FK relationships
- Load data into tables
- Validate row counts

🎯 EPIC 3: Business Use Cases Implementation

User Story 5

As a Business Analyst, I want to identify disease with maximum claims so that I can analyze high-risk diseases.

User Story 6

As a Business Analyst, I want to find subscribers under age 30 so that I can analyze young customer segments.

User Story 7

As a Business Analyst, I want to identify group with maximum subgroups.

User Story 8

As a Business Analyst, I want to find hospital serving the most patients.

User Story 9

As a Business Analyst, I want to identify most subscribed subgroup.

User Story 10

As a Business Analyst, I want to calculate total rejected claims.

User Story 11

As a Business Analyst, I want to identify city with highest number of claims.

User Story 12

As a Business Analyst, I want to compare government vs private policy subscriptions.

User Story 13

As a Business Analyst, I want to calculate average monthly premium paid by subscribers.

User Story 14

As a Business Analyst, I want to identify the most profitable group.

User Story 15

As a Business Analyst, I want to find patients below age 18 admitted for cancer.

User Story 16

As a Business Analyst, I want to identify cashless patients with high hospital charges.

User Story 17

As a Business Analyst, I want to find female patients above 40 who underwent knee surgery.

🎯 EPIC 4: Testing & Validation

User Story 18

As a QA Engineer, I want to validate cleaned data so that it meets quality standards.

Tasks:

- Validate null handling
- Validate duplicate removal
- Check schema consistency

User Story 19

As a QA Engineer, I want to validate output tables so that results are correct.

Tasks:

- Verify SQL query outputs
- Cross-check results with expected logic

🎯 EPIC 5: Deployment & Documentation

User Story 20

As a Data Engineer, I want to push project code to GitHub so that it can be submitted.

Tasks:

- Create GitHub repository
- Push PySpark code
- Upload documentation

User Story 21

As a Data Engineer, I want to prepare a presentation so that I can explain the solution clearly.

Tasks:

- Create slides
- Add architecture diagram
- Include screenshots

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4. Test Cases

Test Case 1

Verify null values are handled correctly

Test Case 2

Verify duplicate records are removed

Test Case 3

Verify data is successfully loaded into Redshift

Test Case 4

Validate each business use case query

Test Case 5

Verify output tables contain expected results

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5. Conclusion

This Jira document outlines the sprint planning, epics, user stories, and test cases required to successfully deliver the healthcare insurance data pipeline project. It ensures structured execution, proper tracking, and alignment with business objectives.